

Chapter 16

A Road Map of Research Methods

16.1 The Road Map

The road map of this book was shown in outline in the Preface and is here shown with more detail in Fig. 16.1 (Research Goals and Research Questions). As stated in the Introduction, design science research iterates over solving design problems and answering knowledge questions. Design problems that need novel treatments are dealt with rationally by the design cycle, which has been treated in Part II. Knowledge questions that require empirical research to answer are dealt with rationally by the empirical cycle, which has been treated in Part IV. Design and empirical research both require theoretical knowledge in the form of conceptual frameworks and theoretical generalizations, which enhance our capability to describe, explain, and predict phenomena and to design artifacts that produce these phenomena. Theoretical frameworks have been treated in Part III.

The outcome of the design cycle is a validated artifact design, not an implementation in the intended social context of use. Implementation in the social context of use is transfer of technology to a stakeholder context of which the original designers of the technology are not part. Technology transfer is not part of design science research but may be a sequel to it. Even technical action research (TAR), in which the newly designed artifact is tested by using it to treat real-world problems, is not technology transfer. Technology transfer is adoption of the artifact by stakeholders without involvement of the design science researcher.

The empirical cycle goes beyond research design by executing the design and analyzing the results. The results are then used to answer the knowledge questions that triggered the empirical cycle.

The different tasks in the empirical cycle have been discussed at length in Part IV, and Appendix B summarizes the cycle in the form of a checklist.

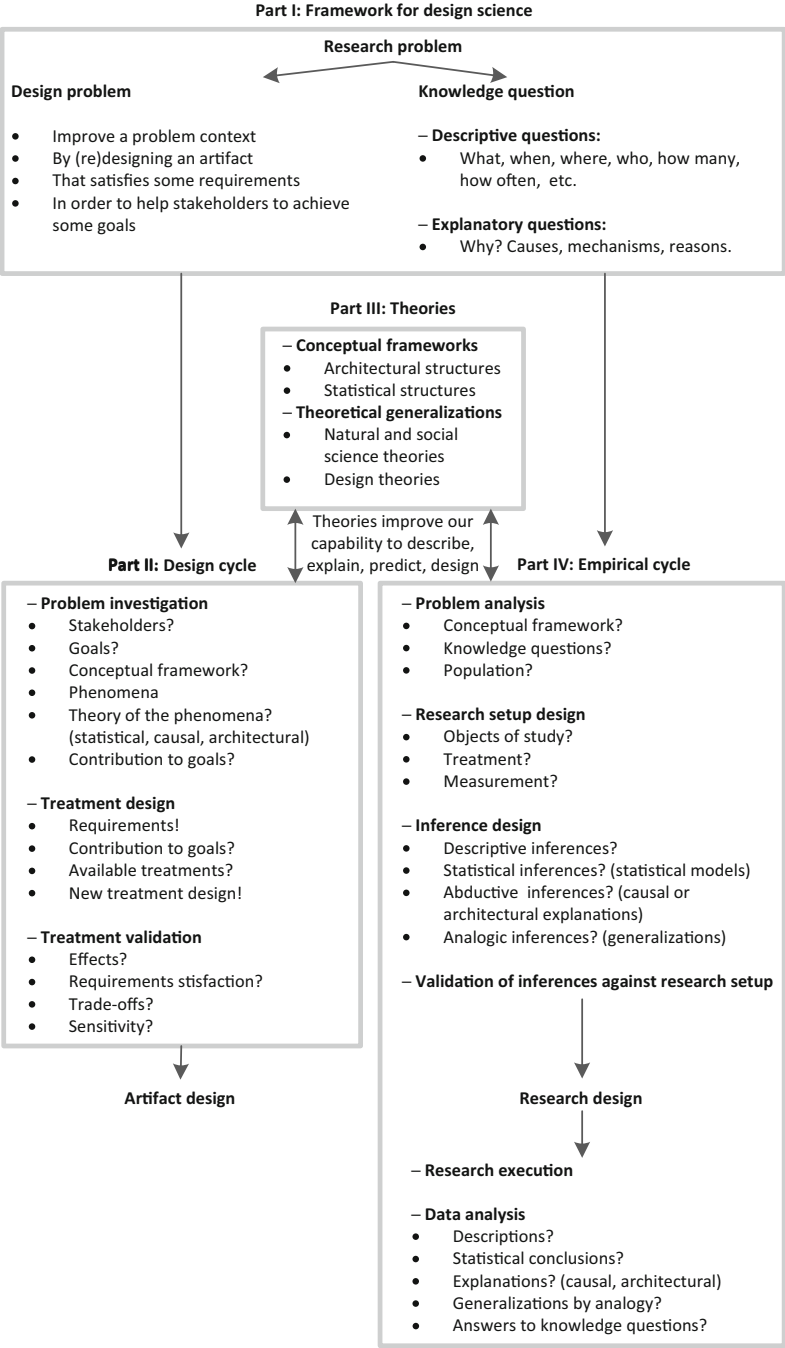


Fig. 16.1 Summary of the design science approach of this book

16.2 Four Empirical Research Methods

In this final part of the book, we show how the checklist can be used to construct four different research methods:

- In *observational case studies*, the researcher studies individual cases to investigate how phenomena in the case are produced by the architecture of the case. The researcher performs no intervention. Observational case studies are done in the field, or at least they are based on information produced in the field. Observational case studies are useful for doing implementation evaluations and problem investigations. They cannot be used to validate new technology because in validation research, by definition, validation models rather than real-world implementations are studied.
- In *single-case mechanism experiments*, the researcher studies individual cases, just as in observational cases studies, to investigate how phenomena in the case are produced by the architecture of the case. But in single-case mechanism experiments, the researcher intervenes, i.e., experiments, with the case. Single-case mechanism experiments are often done in the laboratory, for example, to test an artifact prototype or to simulate real-world phenomena, but they can also be done in the field, for example, to investigate a real-world implementation. Single-case mechanism experiments are useful for validating new technology, for evaluating implementations, and for investigating problems in the field.
- In *TAR*, the researcher experiments with single cases, just as in single-case mechanism experiments. But in contrast to single-case mechanism experiments, this is not only done to answer a knowledge question, but also to help a client. This is useful when validating new technology under real-world conditions. TAR is always done in the field.
- In *statistical difference-making experiments*, the researcher gives two samples of population elements different treatments in order to find out if the difference in treatments causes a difference, on the average, in this population. This can be done in the laboratory or in the field. Statistical difference-making experiments are useful to evaluate implementations in the field, to learn more about problems in the field, and to validate new technology on a large scale in the lab or in the field.

There are additional research methods, not discussed in the chapters that follow. For example, *survey research* is useful for implementation evaluation and problem research but is not discussed here. *Expert opinion* is an effective way to validate new artifact designs at an early stage but is not discussed here. Other important research methods not treated here are *systematic literature surveys* and *meta-analysis*.

The four methods discussed here have many variants too. Observational case studies can be done for cases with a simple or complex nested structure, and it can be done for one or more cases in series. Single-case mechanism experiments can be done with software prototypes, role-play with students, simulations of real-world phenomena, etc. TAR can be done with various levels of involvement

of the researcher. Statistical difference-making experiments have many variants, depending on the number of treatments and samples and on the sampling methods. In the chapters to come, we discuss only a few illustrative examples and show how these examples can be reconstructed using the checklist for empirical research. This should provide enough inspiration for you to construct your own research method based on the checklist.

16.3 One Checklist

The checklist is a list of possible choices but does not prescribe how these choices are to be made. The start of the checklist contains questions to position yourself in the goal structure of design science projects that we discussed earlier in Chap. 2. The checklist then goes to ask what the research problem is, how it is to be investigated, how valid this design is for this problem, etc.

To keep in touch with the top-level structure of your research, it is important to ask yourself early on how your research is positioned along the following four dimensions. These dimensions correspond to major choices in the checklist and can be viewed as a high-level version of the checklist. Two of the dimensions are shown in Fig. 16.2:

- **What is studied.** Do you want to study single cases in-depth, or do you want to study statistics of samples? In other words, do you want to know by what mechanisms single phenomena were produced, or do you want to know what the average value of a variable in a sample is, to compare it with other averages? To answer this question, you should be aware of what the population of interest is and what its elements are.

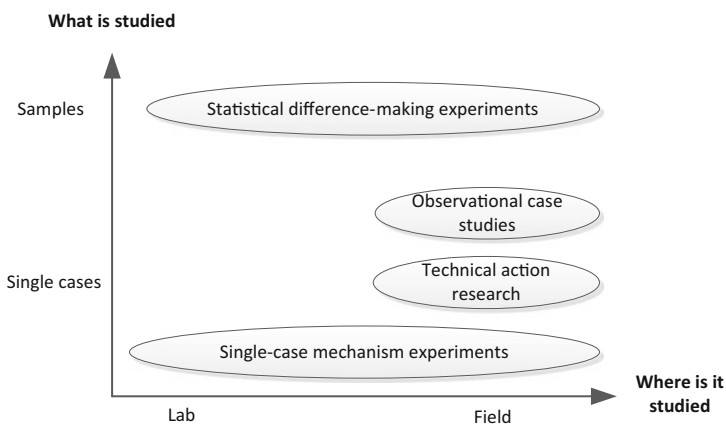


Fig. 16.2 Two dimensions along which to classify research methods

If you are studying a single element at a time, then you are studying single cases. Typically, you want to know how phenomena in the case are produced, and you will generalize by analogy to similar cases, which are the other elements of the population. Case-based research allows you to focus on systematic similarities among population elements. Explanations of phenomena should be architectural. If you are studying sets of elements at a time, then you are studying samples. Typically, you want to know what sample statistics tell you about distribution parameters such as the population average of the variable. Sample-based research allows you to average out nondeterministic differences among population elements. Explanations of phenomena can be causal.

- **Where it is studied.** Whatever you are studying, you may want to study it in the laboratory or in the field. If the phenomenon of interest cannot be produced in the laboratory, you are restricted to do field studies. This is expensive. The population elements that you study may be heterogeneous, and it may be hard to control the factors that influence these elements.
If the phenomenon of interest can be produced in the laboratory, then you have to specify how to do this and how to control the factors that may influence the objects of study. Laboratory studies can be generalized to other laboratory studies. But if your goal is to learn something about phenomena in the real world, then you also have to think about the (dis)similarity of the laboratory setup with real-world setups.
- **How it is studied.** A third dimension, not shown in Fig. 16.2, is whether or not to intervene in the objects of study. In observational studies, you do not intervene; in experimental studies, you do intervene. Observational case studies, surveys, and focus groups to collect expert opinion are observational studies. Observational studies can support architectural explanations, but to provide support for causal explanations, experimental interventions are needed.
- **Why it is studied.** Your choices made along the above dimensions are motivated by your research goals. Do you want to investigate a real-world problem or evaluate an implementation in the real world? Then you should do field research. Do you want to test a prototype of a newly designed artifact? Then case-based or sample-based experimental research in the lab or field should be done.

Additional considerations in choosing a research design include the available resources and possible risks. Your available resources will constrain what you can do. How much time do you have for the research? What resources would you need for a particular research design, and what is the cost of these resources?

Resources of other stakeholders will play a role too. How much time do they have to wait for your results? How much time would subjects have to participate in the research? Can they afford to spend this time?

Risks must be assessed too. Can stakeholders in the research, such as sponsors or you, afford the risk of failed research? For example, can they, or you, afford to run the risk of doing an experiment that has inconclusive results? And is there a risk of harming people's interests in the research?

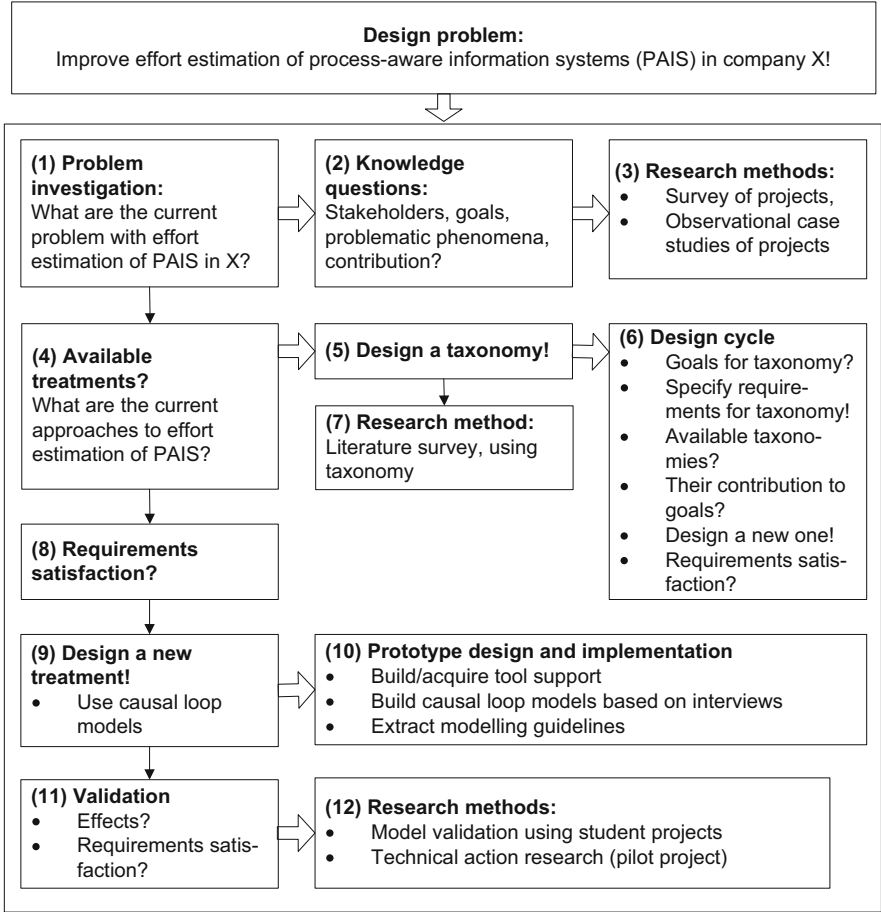


Fig. 16.3 Methodological structure of a PhD thesis. The *hollow arrows* are problem decompositions, the *single arrows* roughly indicate temporal sequence of problem-solving, and the numbers indicate the sequence in which tasks were performed

The following examples illustrate how in different stages of the design cycle different empirical research methods have been selected:

- Mutschler's [2] research goal was to improve effort estimation techniques for process-aware information systems (PAIS). A PAIS is a software system that manages and executes operational business processes. Examples are workflow management systems, case management systems, and enterprise information systems. Figure 16.3 shows the structure of the research project. The boxes represent tasks, double arrows indicate task decompositions, and numbers indicate the sequence in which tasks were performed. The boxes at the left-hand side of the figure are connected by single arrows to indicate that they correspond roughly to the tasks in the design cycle and have been executed from top to bottom.

One of the stakeholders was the large multinational manufacturing company who sponsored Mutschler's research. The problem that triggered the research project was that effort estimations for PAIS development projects were often over budget, which means their effort estimations were inaccurate. To investigate this problem, Mutschler performed two observational field studies, namely, a web-based survey among project managers and some observational case studies of individual projects. They were studied in the field, because the relevant phenomena could not be produced in the laboratory. They were studied observationally, because experimental interventions in the projects would probably destroy the relevant phenomena. The web-based survey is sample based and gives an impression of the extent of the problem. The case studies are case based and give information about the possible reasons and mechanisms why effort estimations are inaccurate. Part of the treatment design task is to make a survey of available effort estimation methods, and for this, Mutschler performed a literature survey. There turned out to be several conflicting taxonomies of effort estimation methods, and before conducting the survey, he constructed a taxonomy based on available taxonomies and his own research goals.

The effort estimation technique designed by Mutschler is based on causal loop models that represent various factors that influence project effort. Altogether, Mutschler made about 20 of these models for different stages of a PAIS development project, based on information collected from project managers.

To validate the models, he performed an experiment in the laboratory with students. Different student projects developed two PAIS each, and the causal loop models were validated by using them to predict the effort that student projects would spend to develop these systems and comparing this with the actual effort spent. This validation was done with students to reduce the risk of burdening project managers in the field with a possibly inaccurate effort estimation technique.

The second validation was done by a real-world project, whose project manager used the causal loop models to estimate project effort. This is an example of TAR. After the project was finished, the estimations were compared with actual effort.

- Gorschek et al. [1] described a design science research project in which researchers worked with practitioners to identify relevant problems experienced by the practitioners and to select one to solve. Their research process follows the design cycle, followed by the first step of implementation, as shown in Table 16.3. In the following, we use the numbering of steps by Gorschek et al.

- (1) Problem investigation was done by observational case studies. One of the problems identified was how to deal with the large volume of requirements, originating from different sources, in market-driven product development, and this problem was selected for treatment.
- (2) Selecting a problem for treatment was done jointly with the industrial stakeholders. This is part of research management, and so it is not included in the engineering cycle.
- (3) Treatment design consisted of a literature survey of the state of the art of requirements engineering, followed by the specification of a candidate solution.
- (4) Validation was done in three iterations. First, the requirements engineering method was used by students in the laboratory. The results motivated an improvement of the method.
- (5) Second, software engineers were asked to express their opinion about the method in focus groups. This is not only a validation but also a part of research management to gain support from important stakeholders. To gain further support, the method was also presented to decision-makers, again for validation as well as for gaining support. These expert opinions led to further improvements of the method, in particular by simplifying it to improve its usability in practice.
- (6) Management gave the go-ahead for the project, and the third validation task was to use the method in two pilot projects. This is an example of two TAR projects. Lessons learned from these projects was used to further improve and finalize the method.
- (7) The final task was to hand over the method to practitioners. This was the start of an implementation process, which is not part of the research project.

Table 16.1 Technology transfer process described by Gorschek et al. [1]

Engineering cycle	Technology transfer model of Gorschek et al. [1]
Problem investigation	1. Identify potential improvement areas based on industry needs
• Observational case study	• Observations
	• Process assessment
	2. Formulate research agenda
	• Identify problems to be treated
	• Formulate problem
Treatment design	3. Formulate candidate solution
	• Survey state of the art
	• Formulate candidate solution
Validation	4. Laboratory validation
• Laboratory experiment	• E.g. by laboratory experiment
	5. Static validation
• Expert opinion	• By expert opinion
	6. Dynamic validation
• Technical action research	• By pilot projects
Implementation	7. Release the solution

Both examples illustrate some frequently made choices in design science research. Problem investigation is often done by observational case studies and surveys. Often the problems cannot be reproduced in the laboratory, and so they must be studied in the field. There is usually no need or no possibility to do experiments to understand the problem, and so the problems are investigated using observational field research. Statistical knowledge about the extent of the problem is obtained by doing a survey; architectural knowledge is obtained by doing observational case studies. A literature study helps to collect what is already known about the problem.

Treatment design always involves making a survey of the state of the art, and this is done by a literature study too.

Treatment validation consists usually of an iteration of validation studies, where early on, laboratory research is done to reduce the cost of research and reduce the risk of harming stakeholders with ineffective treatments. As a preparation for field tests, expert opinion is a cheap and effective method. Prior to releasing a treatment design for implementation in the real world, TAR is used to increase confidence in the validity of the treatment. We recognize here the process of scaling up from the lab to the real world, described in Chap. 7 (Treatment Validation).

References

1. T. Gorschek, C. Wohlin, P. Garre, S. Larsson, A model for technology transfer in practice. *IEEE Softw.* **23**(6), 88–95 (2006)
2. B. Mutschler, Modeling and simulating causal dependencies on process-aware information systems from a cost perspective. Ph.D. thesis, University of Twente, Enschede, January 2008. <http://eprints.eemcs.utwente.nl/11864/>